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EXPERIENCE

ScimerTech

Data Miner

Dec 2023 - Jan 2024, Remote

- Automated **NIST web-book** data scraping, boosting collection efficiency by 85% with advanced extraction techniques.
- Improved spectroscopic analysis accuracy by **72%** through noise reduction, smoothing, and feature extraction using **1-D Convolution** and **Random Forest**.

PROJECTS

Dependency-Aware Transformer Architecture

BTech Paper

Present,

- Improved transformer architecture enhanced for learning **inter word dependency representation** for other downstream tasks.
- Enhanced model explainability compared to BeRT & RoBeRT

Gene-Disease Predictor

ML on Graphs

April 2025,

- Employed **BiANE** for generating node embeddings within a **bipartite graph** aimed at performing a **link prediction** task.

Retina Segmentation

UNet inspired Image Segmentation

Dec 2024,

- Developed UNet architecture for retina image segmentation, effectively delineating retinal structures **medical image analysis**.

CUDA-accelerate

Algorithms for NVIDIA GPUs

Oct 2024,

- Programmed in CUDA to implement **tiled matrix multiplication** and **image processing filters** for enhanced GPU performance.

Nyaaya (Justice) Chatbot

Department of Justice, India

Sept 2024,

- Deployed **vector database** for scalable storage and retrieval, using **HNSW** for optimized search and **RAG**, & **pipelines** for ingesting images, PDFs, and HTML along with **recursive retrieval**.

ACHIEVEMENTS

HackTU 6.0 (MLH Hackathon)

Winner

Feb 2025, TIET Patiala

AudiHealth

- Liquid state machine for detection of vocal pathologies. Accuracy achieved with **BiLSTM 95%**.

Bharatiya Antariksh Hackathon (ISRO)

Finalist

Aug 2024, NRSC Hyderabad

IIRS Spectral Image Classification for Lunar Surface

- **Wavelet-based noise reduction** for lunar geological insights; **82%** accuracy in with **KNN** and **Gaussian clustering**

EDUCATION

Indian Institute of Information Technology (IIIT) Kottayam

BTech Computer Science - AI/DS

GPA: 8.3/10

CERTIFICATION

Google Cloud Platform

Google

Diffusion Model

DeepLearning.ai

N-FAB

IIITK

N-HANS

IIITK

N-LARGE

IIITK

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, Java, Javascript, HTML, CSS, CUDA, Bash

Framework: FastAPI, Apache-Spark, Pytorch, Hugging-Face, TensorFlow, PyGeometric, LangChain, LlamaIndex, StreamLit, MATLAB

Developer Tools: Git, Docker, CMake, MongoDB, MLFlow, Envi, ImageJ

Libraries: Pandas, NumPy, Matplotlib, Keras, NLTK, Seaborn, OpenCV, Spacy, Scikit-Learn, Graph4NLP, Transformer, NetworkX, OpenNLP, Fasttext, DGL

COURSES

- Cloud Computing
- Machine Learning
- Computer Network
- Probability & Statistics
- Optimization for AI/ML
- Graph Machine Learning
- Linear Algebra & Calculus
- Parallel & Distributed Computing

OTHER INTERESTS

- Robotics
- Dynamic & Temporal Graphs
- Federated Learning